

Trainee Management System

The first real project for DTTP trainees. Build it as a genuine internal business application, not as a tutorial project.

01 Project goal

Build a web application to manage the complete DTTP trainee lifecycle.

Registration → Batch Assignment → Training → Attendance
→ Tasks & Assignments → Evaluation → Feedback → Completion

02 Technology stack

BACKEND

Python · FastAPI · SQLAlchemy ·
MySQL · JWT · Pydantic · Pytest

FRONTEND

React · TypeScript · HTML/CSS ·
REST API

TOOLS

VS Code · Git · GitHub · Postman ·
MySQL Workbench · Docker ·
GitHub Actions

03 User roles

ADMIN

Complete system.

- Trainees
- Mentors
- Batches
- Projects
- Users
- Reports

MENTOR

Assigned trainees.

- Attendance
- Tasks
- Assignments
- Reviews
- Feedback
- Performance

TRAINEE

Own activities.

- Profile
- Attendance
- Tasks
- Assignments
- Submissions
- Feedback
- Progress

04 Module 1 — Authentication

Login · Logout · JWT · Role-based access · Password management

ADMIN

MENTOR

TRAINEE

05 Module 2 — Trainee management

Admin can add, view, edit, search, filter, activate/deactivate, assign batch, and assign mentor.

Fields: Trainee ID, Name, Email, Mobile, Address, Education, Joining Date, Batch, Mentor, Status.

06 Module 3 — Batch management

Admin creates batches such as DTTP-2026-01, DTTP-2026-02, DTTP-2026-03.

Fields: Batch Name, Start Date, End Date, Mentor, Capacity, Status.

07 Module 4 — Mentor management

Admin can add, update, assign, view assigned trainees, and activate/deactivate a mentor.

08 Module 5 — Attendance

Mentor marks Present, Absent, Late, or Leave.

Date	Trainee	Status
19-Aug	Rahul	Present
19-Aug	Amit	Absent
19-Aug	Sana	Present

Reports: daily, monthly, individual, attendance percentage.

09 Module 6 — Task management

Fields: Title, Description, Priority, Deadline, Assigned To.

TODO	IN_PROGRESS	SUBMITTED	UNDER_REVIEW	COMPLETED
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10 Module 7 — Assignment and submission

Mentor → Create Assignment → Trainee → Complete Work → Submit → Mentor Review → Feedback

Submission: description, GitHub repository, file/document, submission date, comments.

11 Module 8 — Evaluation

Skill	Score
Python	8/10
FastAPI	7/10
MySQL	8/10
React	7/10
Git	9/10
Problem Solving	8/10
Communication	8/10

12 Module 9 — Feedback

Mentor provides technical, task, and project feedback, plus strengths, areas for improvement, and recommendations. Trainee views feedback from the dashboard.

13 Module 10 — Dashboard

ADMIN

- Total / active trainees
- Total batches and mentors
- Today's attendance
- Pending tasks and reviews

MENTOR

- My trainees
- Today's attendance
- Pending tasks
- Pending submissions
- Performance

TRAINEE

- My profile
- Attendance %
- Pending / completed tasks
- Assignments
- Performance and feedback

14 Database design

Tables: users · roles · trainees · mentors · batches · attendance · tasks · assignments · submissions · evaluations · feedback

```
User → Trainee | Mentor
Batch → Trainees
Mentor → Trainees
Trainee → Attendance, Tasks, Assignments, Submissions, Evaluations, Feedback
```

15 Backend architecture

```
FastAPI
├── Routers   ├── Schemas   ├── Services   └── Models
├── Database ├── Auth       └── Exceptions
└──
    ↓
SQLAlchemy → MySQL
```

```
backend/
├── app/ (main.py, config, database, models, schemas,
│       routers, services, auth, exceptions)
├── tests/
├── requirements.txt
├── .env          ← never commit secrets
└── README.md
```

16 Frontend architecture

React: Pages · Components · Layouts · Services · Hooks · Routes · Types · Utils

Screens: Login, Dashboard, Trainees, Trainee Details, Batches, Mentors, Attendance, Tasks, Assignments, Submissions, Evaluation, Feedback, Profile.

17 First MVP

Do not build all modules at once.

```
Login → Admin Dashboard → Trainee Management → Batch Management
```

```
POST /api/auth/login
POST /api/trainees
GET /api/trainees
GET /api/trainees/{id}
PUT /api/trainees/{id}
DELETE /api/trainees/{id}
POST /api/batches
GET /api/batches
GET /api/batches/{id}
PUT /api/batches/{id}
```

18 Development sprints

Sprint	Work
1	GitHub repo, backend, frontend, MySQL, structure, README
2	Trainee model, CRUD APIs, validation, Postman
3	React login, dashboard, trainee list, add/edit, API integration
4	Login, JWT, roles, protected APIs
5	Batch, mentor, assignment
6	Mark attendance, reports, percentage
7	Tasks, assign, submission, review
8	Evaluation, feedback, progress
9	Unit, API, frontend tests, bug fixing
10	GitHub → Actions → Tests → Build → Docker → Deployment

19 First real-time requirement

As an Admin, I want to register DTTP trainees and assign them to a batch so that I can maintain a centralized trainee database.

- | | |
|---|---|
| ☐ Admin can log in | ☐ Admin can deactivate a trainee |
| ☐ Admin can create a trainee | ☐ Admin can assign a batch |
| ☐ Required fields are validated | ☐ Data is stored in MySQL |
| ☐ Email cannot be duplicated | ☐ APIs are tested |
| ☐ Admin can view trainees | ☐ Frontend works |
| ☐ Admin can search trainees | ☐ Code is committed through Git |
| ☐ Admin can edit trainee details | ☐ Pull request is reviewed by mentor |

20 Trainee development workflow

```
Requirement → Issue → Feature Branch → Design → Code → Test  
→ Commit → Push → Pull Request → Mentor Review → Fix → Approval → Merge
```

The first project should teach how real software teams work, not merely how to write Python.

First milestone: build and deploy a working TMS MVP using Python + FastAPI + MySQL + React, with GitHub-based code review.